DATABASE MANAGEMENT SYSTEM (ENCS205)

Lab Report Submitted to

K. R. Mangalam University

for

K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

Bachelor of Technology
in
Computer Science and Engineering
Submitted by

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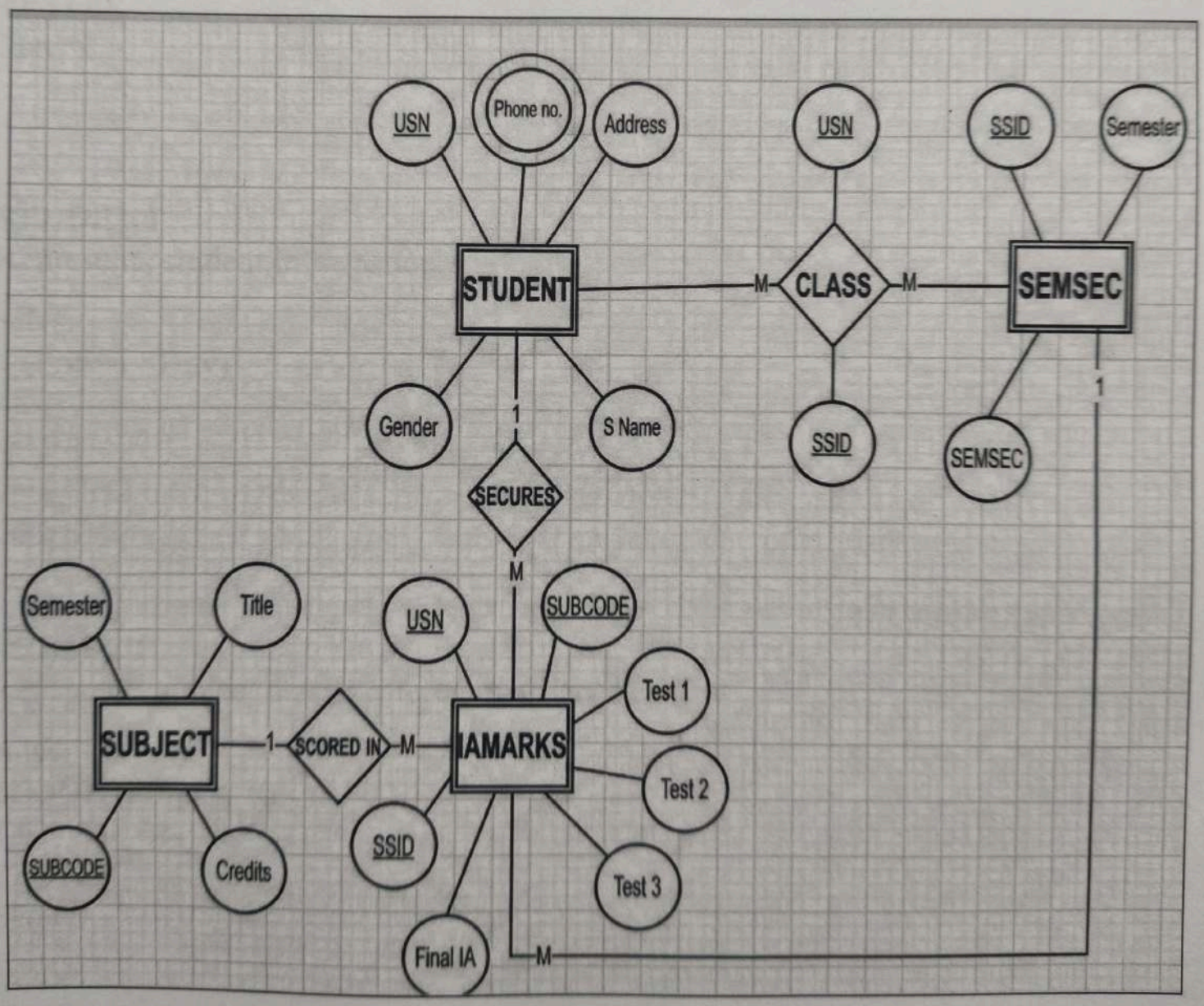
Experiment No: 1

Aim:

To design an Entity Relationship (ER) diagram for the College database and convert the ER diagram into relational tables.

E R diagram:



ENTITIES AND ATTRIBUTES

1. SEMSEC

Represents semester and section details.

Attributes:

- SSID (Primary Key)
- SEMESTER
- SECTION

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2. STUDENT

Represents student information.

Attributes:

- USN (Primary Key)
- SNAME
- GENDER
- PHONE_NO
- ADDRESS

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3. SUBJECT

Represents subject details offered in a semester.

Attributes:

- SUBCODE (Primary Key)
- TITLE
- SEMESTER
- CREDITS

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4. IAMARKS

Represents internal assessment marks secured by students in subjects.

Attributes:
- USN (Foreign Key)
- SUBCODE (Foreign Key)
- SSID (Foreign Key)
- TEST1
- TEST2
- TEST3
- FINALIA (Derived Attribute)

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5. CLASS

Represents a class allotted to students.

Attributes:
- USN
- SSID

---

6. SECURES

Represents the marks secured by a student.

Attributes:
- USN
- SUBCODE

---

7. SCORED_IN

Represents the subject in which marks are scored.

Attributes:
- SUBCODE
- USN

Relationships:

1. **CLASS**
   Represents the relationship between SEMSEC and STUDENT.
   One SEMSEC can have many STUDENTS, but each STUDENT belongs to one SEMSEC.
   Cardinality: 1 : N

2. **SECURES**
   Represents the relationship between STUDENT and IAMARKS.
   One STUDENT can secure many IAMARKS records.
   Each IAMARKS record belongs to one STUDENT.
   Cardinality: 1 : N

3. **SCORED_IN**
   Represents the relationship between SUBJECT and IAMARKS.
   One SUBJECT can have many IAMARKS records.
   Each IAMARKS record belongs to one SUBJECT.
   Cardinality: 1 : N

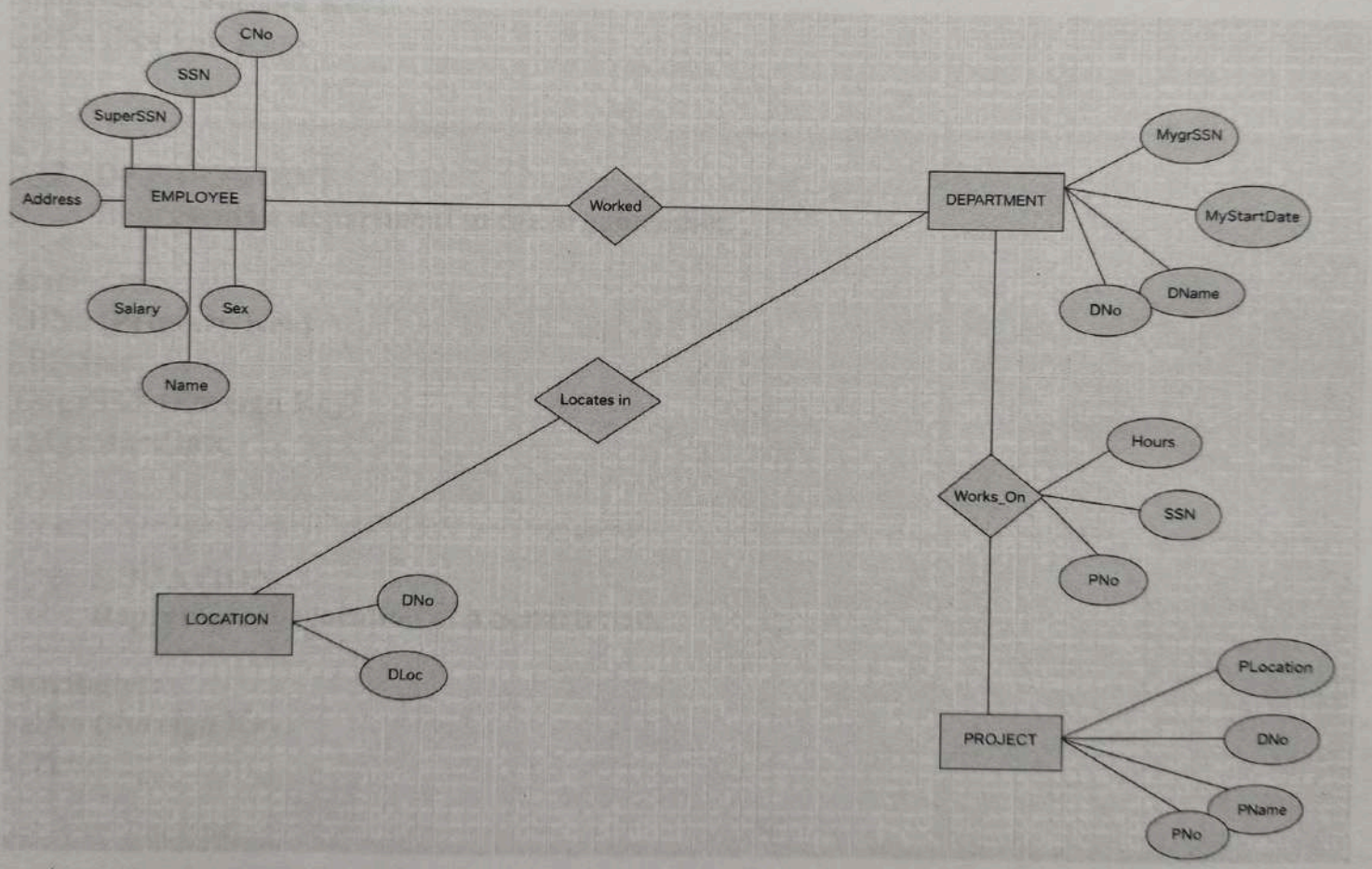
Conclusion: This ER diagram clearly explains how students are related to classes, semesters, subjects, and marks. It helps in designing a proper database structure for managing academic records efficiently.

Experiment No.2

Aim:

To design an Entity Relationship (ER) diagram for the Company database and convert it into relational tables.

E R diagram:



Entities and Attributes:

1. **EMPLOYEE**
   Represents an employee working in the organization.

Attributes:
- **SSN (Primary Key)**
- **Name**
- **Address**
- **Salary**
- **Sex**
- **SuperSSN (Foreign Key)**
- **DNo (Foreign Key)**

---

2. **DEPARTMENT**
   Represents a department in the organization.

Attributes:
- **DNo (Primary Key)**
- **DName**
- **MgrSSN (Foreign Key)**
- **MgrStartDate**

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3. **LOCATION**
   Represents the location of a department.

Attributes:
- **DNo (Foreign Key)**
- **DLoc**

---

4. **PROJECT**
   Represents projects handled by departments.

Attributes:
- **PNo (Primary Key)**
- **PName**
- **PLocation**
- **DNo (Foreign Key)**

Relationship:

1. **EMPLOYEE – DEPARTMENT**
   Relationship: Worked
   An employee works in one department, and a department can have many employees.
   Cardinality: 1 : N

---

2. **DEPARTMENT – PROJECT**
   Relationship: Works On
   A department controls many projects, and each project is associated with one department.
   Cardinality: 1 : N

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3. **EMPLOYEE – PROJECT**
   Relationship: Works On
   An employee can work on multiple projects, and a project can have multiple employees.
   The relationship stores the attribute Hours.
   Cardinality: M : N

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4. **DEPARTMENT – LOCATION**
   Relationship: Located In
   A department can be located in multiple locations, and each location belongs to one department.
   Cardinality: 1 : N

Conclusion:

This ER diagram clearly illustrates the relationships between employees, departments, projects, and locations in a company. It serves as a strong foundation for designing a well-structured and efficient company database system.

Experiment No.3

Aim:

To design and implement a Customer–Sale database schema using SQL by creating tables with appropriate integrity constraints, inserting records, and performing various queries and views to retrieve bill details, customer purchase information, product sales, and daily sales analysis.

1. Create the tables with the appropriate integrity constraints.

```sql
CREATE DATABASE IF NOT EXISTS assignment3;
USE assignment3;
CREATE TABLE Customer (
    cust_id INT PRIMARY KEY,
    cust_name VARCHAR(100) NOT NULL
);
CREATE TABLE Item (
    item_id INT PRIMARY KEY,
    item_name VARCHAR(100) NOT NULL,
    price INT NOT NULL CHECK (price > 0)
);
CREATE TABLE Sale (
    bill_no INT PRIMARY KEY,
    bill_date DATE NOT NULL,
    cust_id INT,
    item_id INT,
    qty_sold INT CHECK (qty_sold > 0),
    FOREIGN KEY (cust_id) REFERENCES Customer(cust_id),
    FOREIGN KEY (item_id) REFERENCES Item(item_id)
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Tables_in_assignment3
▶	bill_view
	customer
	item
	sale
	weekly_sales_view

2. Insert around 10 records in each of the tables.

```
INSERT INTO Customer VALUES
(1,'Yashi'),
(2,'Aani'),
(3,'Aahan'),
(4,'Priya'),
(5,'Arjun'),
(6,'Neha'),
(7,'Karan'),
(8,'Sneha'),
(9,'Rohit'),
(10,'Meena');
```

```
INSERT INTO Item VALUES
(101,'Pen',50),
(102,'Notebook',120),
(103,'Bag',800),
(104,'Bottle',250),
(105,'Shoes',1500),
(106,'Watch',2200),
(107,'T-shirt',500),
(108,'Mouse',700),
(109,'Keyboard',900),
(110,'Book',300);
```

```
INSERT INTO Sale VALUES
(1001,CURDATE(),1,103,1),
(1002,CURDATE(),2,101,5),
(1003,CURDATE(),3,105,2),
(1004,CURDATE(),4,104,3),
(1005,CURDATE(),5,110,4),
(1006,CURDATE()-INTERVAL 1 DAY,6,106,1),
(1007,CURDATE()-INTERVAL 2 DAY,7,102,6),
(1008,CURDATE(),8,108,2),
(1009,CURDATE()-INTERVAL 3 DAY,9,109,1),
(1010,CURDATE(),10,107,3);
```

1009	2026-02-20	9	109	1
1010	2026-02-23	10	107	3
NULL	NULL	NULL	NULL	NULL

3. List all the bills for the current date with the customer names and item numbers.

```
SELECT s.bill_no,
       s.bill_date,
       c.cust_name,
       s.item_id
FROM Sale s
JOIN Customer c ON s.cust_id = c.cust_id
WHERE s.bill_date = CURDATE();
```

Result Grid | Filter Rows: | Export:

bill_no	bill_date	cust_name	item_id
1001	2026-02-23	Yashi	103
1002	2026-02-23	Aani	101
1003	2026-02-23	Aahan	105
1004	2026-02-23	Priya	104
1005	2026-02-23	Arjun	110
1008	2026-02-23	Sneha	108
1010	2026-02-23	Meena	107

4. List the total Bill details with the quantity sold, price of the item and the final amount.

```
SELECT s.bill_no,
       s.qty_sold,
       i.price,
       (s.qty_sold * i.price) AS final_amount
FROM Sale s
JOIN Item i ON s.item_id = i.item_id;
SELECT bill no, bill date FROM Sale;
```

Result Grid | Filter Rows:

bill_no	bill_date
1001	2026-02-23
1002	2026-02-23
1003	2026-02-23
1004	2026-02-23
1005	2026-02-23
1006	2026-02-22
1007	2026-02-21
1008	2026-02-23
1009	2026-02-20
1010	2026-02-23
NULL	NULL

5. List the details of the customer who have bought a product which has a price>200.

```
SELECT DISTINCT c.cust_id,
       c.cust_name
FROM Sale s
JOIN Customer c ON s.cust_id = c.cust_id
JOIN Item i ON s.item_id = i.item_id
WHERE i.price > 200;
```

Result Grid | Filter Rows:

cust_id	cust_name
1	Yashi
4	Priya
3	Aahan
6	Neha
10	Meena
8	Sneha
9	Rohit
5	Arjun

6. Give a count of how many products have been bought by each customer.

```
SELECT c.cust_id,
       c.cust_name,
       SUM(s.qty_sold) AS total_products_bought
FROM Sale s
JOIN Customer c ON s.cust_id = c.cust_id
GROUP BY c.cust_id, c.cust_name;
```

Result Grid | Filter Rows: | Export:

cust_id	cust_name	total_products_bought
1	Yashi	1
2	Aani	5
3	Aahan	2
4	Priya	3
5	Arjun	4
6	Neha	1
7	Karan	6
8	Sneha	2
9	Rohit	1
10	Meena	3

7. Give a list of products bought by a customer having cust_id as 5.

```
SELECT i.item_id,
       i.item_name,
       i.price,
       s.qty_sold
FROM Sale s
JOIN Item i ON s.item_id = i.item_id
WHERE s.cust_id = 5;
```

Result Grid | Filter Rows: | E

item_id	item_name	price	qty_sold
110	Book	300	4

8. List the item details which are sold as of today.

```
SELECT DISTINCT i.item_id,
       i.item_name,
       i.price
FROM Sale s
JOIN Item i ON s.item_id = i.item_id
WHERE s.bill_date = CURDATE();
```

Result Grid | Filter Rows:

item_id	item_name	price
103	Bag	800
101	Pen	50
105	Shoes	1500
104	Bottle	250
110	Book	300
108	Mouse	700
107	T-shirt	500

9. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.

```
CREATE VIEW Bill_View AS
SELECT s.bill_no,
       s.bill_date,
       s.cust_id,
       s.item_id,
       i.price,
       s.qty_sold,
       (s.qty_sold * i.price) AS amount
FROM Sale s
JOIN Item i ON s.item_id = i.item_id;
SELECT * FROM Bill_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

bill_no	bill_date	cust_id	item_id	price	qty_sold	amount
1001	2026-02-23	1	103	800	1	800
1002	2026-02-23	2	101	50	5	250
1003	2026-02-23	3	105	1500	2	3000
1004	2026-02-23	4	104	250	3	750
1005	2026-02-23	5	110	300	4	1200
1006	2026-02-22	6	106	2200	1	2200
1007	2026-02-21	7	102	120	6	720
1008	2026-02-23	8	108	700	2	1400
1009	2026-02-20	9	109	900	1	900
1010	2026-02-23	10	107	500	3	1500

10. Create a view which lists the daily sales date wise for the last one week

```
CREATE VIEW Weekly_Sales_View AS
SELECT s.bill_date,
       SUM(s.qty_sold * i.price) AS daily_total_sales
FROM Sale s
JOIN Item i ON s.item_id = i.item_id
WHERE s.bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY s.bill_date;
SELECT * FROM Weekly_Sales_View;
```

Result Grid | Filter Rows:

bill_date	daily_total_sales
2026-02-23	8900
2026-02-22	2200
2026-02-21	720